<u>Thm</u>
- $\dim S_k \le k$
- $S_{n-1} = S_{n-2} \implies \dim S \le n-2$

<u>Proof sketch</u>

- <u>$\dim S_k \le k$</u> : Write $S_k = \bigcup_{i \geq 1} S_{k,i}$

$$S_{k,i} = \left\{ y \in Y \;\middle|\; \begin{array}{l} d_{GH}\big(B_r(y), B_r(\overbrace{(0,x^*)}^{\in \mathbb{R}^{k+1} \times C(X)})\big) \geq i^{-1} r \\ \forall\, 0 < r \le i^{-1}, \quad \begin{array}{l} 0 \in \mathbb{R}^{k+1} \\ x^* \in C(X) \\ \text{vertex} \end{array} \end{array} \right\}$$

we show $\dim S_{k,i} \le k$.

Suppose not, i.e. $\mathcal{H}^{k'}(S_{k,i}) > 0$ for some $k' > k$.

Let $y \in D_{k'}(S_{k,i})$ be a $k'$-density point

Let $Y_y$ = tangent cone at $y$ obtained as a limit of $(Y, y, r_\ell^{-1} d_Y)$ s.t.

$$2^{-k'} \le \lim_{\ell \to \infty} \frac{\mathcal{H}^{k'}_\infty(S_{k,i} \cap B_{r_\ell}(y))}{\omega_{k'} r_\ell^{k'}}$$

As $\ell \to \infty$, the sets

$S_{k,i} \cap \overline{B_{r_\ell}(y)}$ scaled up by $r_\ell^{-1}$

converge to set contained in $S_{k,i}(Y_y) \cap \overline{B_1(y_\infty)}$

$$\implies \mathcal{H}^{k'}(S_{k,i}(Y_y) \cap \overline{B_1(y_\infty)}) > 0$$

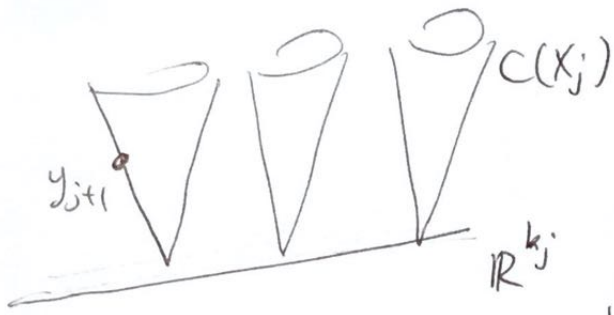
Can repeat this $\rightsquigarrow$ get sequence of iterated tangent cones $Y_{y_j}$ s.t.

$y_j$ is an arbitrary pt. of $k'$-density in $Y_{y_{j-1}}$

and we get $\mathcal{H}^{k'}(S_{k,i}(Y_{y_j}) \cap \overline{B_1(y_{j,\infty})}) > 0$

<u>Note</u>: • If $Y_{y_j} = \mathbb{R}^{k_j} \times C(X_j)$, then $k_j \leq k$, since otherwise $S_{k,i}(Y_{y_j}) = \emptyset$

• We can ensure $y_{j+1} \notin \mathbb{R}^{k_j} \times \{x_j^*\}$




by splitting thm $k_{j+1} \geq k_j + 1$

$\Longrightarrow$ we have contradiction for $j > k$. $\square$

• <u>$S = S_{n-2}$</u>: if $y \in S_{n-1} \setminus S_{n-2}$

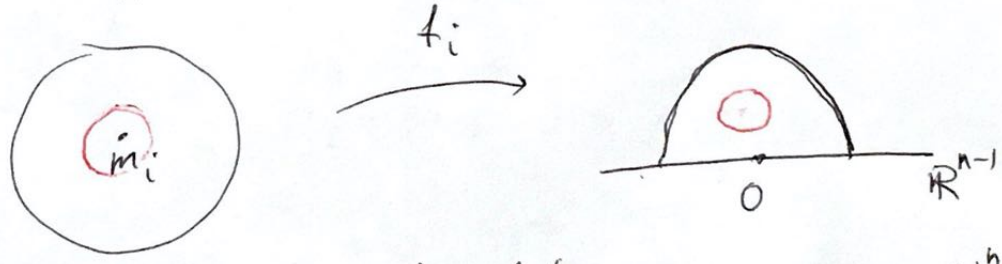
$\Longrightarrow \exists$ tan. cone $Y_y = \mathbb{R}^{n-1} \times C(X)$

but $Y_y \neq \mathbb{R}^n$, so $C(X) = \mathbb{R}_+$

i.e. $Y_y = H^n = \mathbb{R}^{n-1} \times \mathbb{R}_+$ half space

We would have balls $B_1(m_i) \xrightarrow{GH} B_1(0) \cap H^n$

can arrange to have continuous GH - approximations




$B_1(m_i)$ s.t. $f_i(\partial B_1(m_i)) \subset \partial B_1(0) \cap H^n$

doubling along boundaries $\partial B_1(m_i)$, $\partial B_1(0) \cap H^n$

$$\widetilde{B_1(m_i)} \xrightarrow{\tilde{f}_i} \text{top. disk}$$

closed mfld

mod 2 degree of $\tilde{f}_i = 0$

as before can find $x \in B_{1/2}(m_i)$ s.t.

$$f_i^{-1}(f_i(x)) = \{x\} \text{ for } i \gg 1$$

i.e. mod 2 degree is 1. $\square$

## Two sided Ricci bounds

Suppose $M_i^n$ satisfy $|Ric_{M_i^n}| \leq n-1$,

$$vol(B_1(m_i)) \geq v > 0.$$

$$(M_i^n, m_i) \xrightarrow{GH} (Y, \bar{y})$$

<u>Thm</u> $\exists\, \varepsilon = \varepsilon(n) > 0$ s.t. $\mathcal{R}_\varepsilon = \mathcal{R}$ and $\mathcal{R}$ is a $C^{1,\alpha}$ Riemannian mfld.

In particular $S = Y \setminus \mathcal{R}$ is closed.

<u>Proof</u> Let $y \in \mathcal{R}_\varepsilon$. Then every tan. cone $Y_y$ at $y$ satisfies

$$d_{GH}(B_1(y_\infty), B_1(0)) < \varepsilon, \quad 0 \in \mathbb{R}^n$$

$\Rightarrow \exists\, r_y > 0$ s.t. $d_{GH}(B_{r_y}(y), B_{r_y}(0)) < \varepsilon r_y$

$\Rightarrow$ if $p_i \in M_i^n$ s.t. $p_i \to y$ then

$$d_{GH}(B_{r_y}(p_i), B_{r_y}(0)) < \varepsilon r_y \text{ for } i \gg 1$$

Rescale by $r_y^{-1}$ to get $\tilde{p}_i \in \tilde{M}_i^n$ :

$$|Ric_{\tilde{M}_i}| \leq (n-1) r_y^2$$

$$d_{GH}(B_1(\tilde{p}_i), B_1(0)) < \varepsilon \quad \text{for } i \gg 1.$$

Vol. convergence $\Longrightarrow$

$$Vol(B_1(\tilde{p}_i)) \geq (1 - \Psi(\varepsilon, r_y | n)) Vol(B_1(0))$$

by Anderson's thm if $\varepsilon, r_y$ are suff. small,

then $\|(\tilde{M}_i, \tilde{g}_i, \tilde{p}_i)\|_{C^{1,\alpha}, r} \leq 2$

for some $r = r(n)$ (for $i \gg 1$)

letting $i \to \infty$ we get $C^{1,\alpha}$ chart on

$$B_r(\tilde{y}) \subset \tilde{Y}.$$

<u>Consequence</u>: $\forall y \in \mathcal{R}_\varepsilon \subset Y$ $\exists$ $C^{1,\alpha}$ chart on neighborhood $B_{r_y \cdot r(n)}(y)$

It follows that $\forall y' \in B_{r_y \cdot r(n)}(y)$ $Y_{y'} = \mathbb{R}^n$

i.e. $B_{r_y \cdot r(n)}(y) \subset \mathcal{R}$ and $\mathcal{R}$ is $C^{1,\alpha}$ Riem. mfld. $\square$

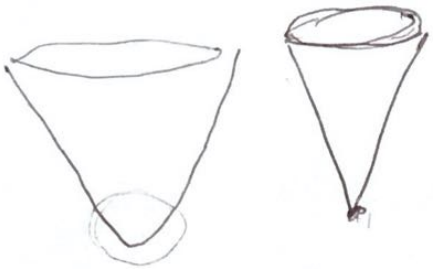
<u>Theorem</u> If $|Ric_{M_i^n}| \leq n-1$, then

$\mathcal{S}_{n-4} = \mathcal{S}$, i.e. $\dim \mathcal{S} \leq n-4$

(Cheeger - Colding - Tian in Kähler case
Cheeger - Naber in general)

Key point is to rule out tangent cones of form $\mathbb{R}^{n-2} \times C(S^1_\gamma)$ ← circle of length $\gamma \in (0, 2\pi)$

e.g. in 2-dim if ball is GH-close to



then by Gauss-Bonnet, curvature must concentrate near tip (cannot happen if $|Ric| \leq 1$)